

Homework 8 - Results

Unformatted Output

Running Task 1: CPU Reduction
CPU-Avg-Time : 30.73680401ms

Running Task 2: GPU cascaded
host 10485760.00 gpu 10485760.00
Reductions match.
GPU-Avg-Time atomic Reduction: 0.22468901ms

Running Task 3: GPU Harris
Number of SMs: 40
Max Threads per SM: 1024
host 10485760.00 gpu 10485760.00
Reductions match.
GPU-Avg-Time Harris: 0.16854501ms

Performance Summary (Formatted Output)

Implementation	Average Time	Speedup vs CPU
CPU Reduction (Sequential)	30.74 ms	1.0x
GPU Atomic Cascaded (HW7)	0.22 ms	~139.7x
GPU Harris Cascaded + threadfence	0.17 ms	~180.8x